ZCAS University

SCHOOL OF COMPUTING TECHNOLOGY AND APPLIED SCIENCES

CCS3142 MOBILE APPLICATION DEVELOPMENT

MID SEMESTER EXAMINATION

THURSDAY, 26 OCTOBER 2023

08:30 - 11:30 HOURS

TIME ALLOWED : WRITING – THREE HOURS : READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

NOTE: Your answers should NOT be brief. Give detailed essay style reasoned answers with structured paragraphs and preferably with headings.

Mobile app development is the process of creating software for mobile devices, such as smartphones and tablets. Mobile apps are typically downloaded from app stores, such as the Apple App Store or Google Play Store.

There are two main types of mobile apps: native and cross-platform. Native apps are developed specifically for one mobile platform, such as iOS or Android. Cross-platform apps are developed using a single codebase that can be deployed to multiple mobile platforms.

Native apps are generally considered to be the best performing and most user-friendly type of mobile app. However, they can be more expensive and time-consuming to develop than cross-platform apps.

Cross-platform apps are a good option for businesses that want to develop a mobile app for multiple platforms without having to invest in separate development teams for each platform. However, they may not perform as well as native apps, and they may not have access to all of the features of each platform.

a) Explain the components of the Android development environment, including Android Studio, the Android SDK, and the emulator. Describe how developers can set up their development environment. **(10 marks)**

b) Describe the different states of an Android app, such as the "Active," "Paused," and "Stopped" states, and explain how they impact app behavior and resource management. **(10 marks)**

c) Discuss the importance of accessibility in Android app design and provide guidance on making apps more accessible to individuals with different abilities. **(10 marks)**

d) Compare and contrast popular architectural patterns like MVC (Model-View-Controller) and MVVM (Model-View-ViewModel) in the context of Android development. Explain their advantages and disadvantages. **(10 marks)**

[Total: 40 Marks]

SECTION B: Attempt any THREE questions in this section

Question 2

a) Mobile applications can be categorized into various types based on their purpose, functionality, and target audience. Identify and describe in detail three categories into which mobile apps can placed. **(6 marks)**

b) Mobile platform architectures refer to the underlying software and hardware structures that power mobile devices, such as smartphones and tablets. Write short notes on the following.

i) Monolithic Architecture:
ii) Microservices Architecture:
iii) Layered Architecture: **(6 marks)**

c) Navigation in Flutter is the process of moving from one screen to another. What are the benefits of using navigation. Write a Flutter dart code segment that will navigate from one screen to the other. **(8 marks)**

[Total: 20 Marks]

Question 3

a) Explore the relationship between user interface design, user experience, and user satisfaction on mobile devices. **(6 marks)**

b) Design patterns play a critical role in developing maintainable and efficient mobile applications. Discuss the role the Singleton and factory pattern play in making mobile applications efficient. **(6 marks)**

c) Critically study the code below. Explain the function of the code. **(8 marks)**

```
1. Future<void> fetchData() async {
2.         final response = await
   http.get(Uri.parse('https://api.example.com/data'));
3.         if (response.statusCode == 200) {
4.               print('Response data: ${response.body}');
5.         } else {
6.         print('Request failed with status: ${response.statusCode}');
7.         }
8. }
```

[Total: 20 Marks]

Question 4

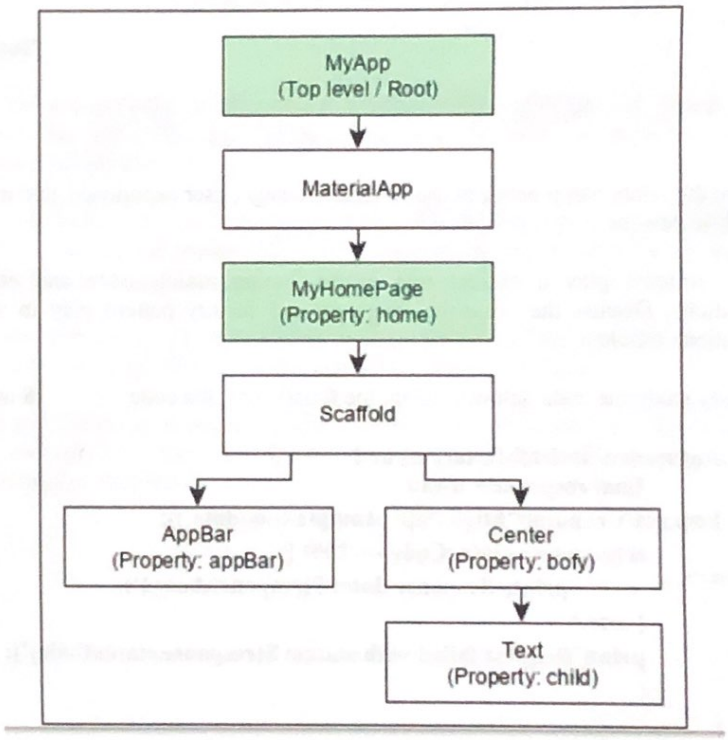
a) Below are widgets that are commonly used in mobile application interfaces. Write Flutter dart code segment outlines that will create these widgets.

appBar
TextFiled
Scafold
Padding

(8 marks)

b) There are many mobile development frameworks available, which provide excellent features to develop mobile applications. Explain why you would choose **Flutter** over the **Java** framework. **(6 marks)**

c) Widgets are basically user interface components used to create the user interface of the application. Explain the function of the widgets shown in the diagram below. **(6 marks)**



[Total: 20 Marks]

Question 5

a) What is the key difference between a StatefulWidget and a StatelessWidget in Flutter, and in what scenarios would you choose one over the other when building UI elements?

(6 marks)

b) Core Android application components are fundamental building blocks that define the structure and behavior of Android applications. These components are essential for creating interactive and functional apps. Identify and describe any three main core android application components.

(6 marks)

c) Study the code below. You are required to draw the user interface that the code will produce:

(8 marks)

```
@override
 Widget build(BuildContext context) {
  return Scaffold(
   appBar: AppBar(),
   drawer: Drawer(
    child: ListView(
     children: [
      UserAccountsDrawerHeader(
       accountName: Text("AppMaking.co"),
       accountEmail: Text("sundar@appmaking.co"),
       currentAccountPicture: CircleAvatar(
        backgroundImage: NetworkImage(
         "https://appmaking.co/wp-content/uploads/2021/08/appmaking-logo-colored.png"),
       ),
       decoration: BoxDecoration(
        image: DecorationImage(
         image: NetworkImage(
          "https://appmaking.co/wp-content/uploads/2021/08/android-drawer-bg.jpeg",
         ),
         fit: BoxFit.fill,
        ),
       ),
       otherAccountsPictures: [
        CircleAvatar(
         backgroundColor: Colors.white,
         backgroundImage: NetworkImage(
          "https://randomuser.me/api/portraits/women/74.jpg"),
        ),
        CircleAvatar(
```

```
30.              backgroundColor: Colors.white,
31.              backgroundImage: NetworkImage(
32.                "https://randomuser.me/api/portraits/men/47.jpg"),
33.            ),
34.          ],
35.        ),
36.        ListTile(
37.          leading: Icon(Icons.home),
38.          title: Text("Home"),
39.          onTap: () {},
40.        ),
41.        ListTile(
42.          leading: Icon(Icons.account_box),
43.          title: Text("About"),
44.          onTap: () {},
45.        ),
46.        ListTile(
47.          leading: Icon(Icons.grid_3x3_outlined),
48.          title: Text("Products"),
49.          onTap: () {},
50.        ),
51.        ListTile(
52.          leading: Icon(Icons.contact_mail),
53.          title: Text("Contact"),
54.          onTap: () {},
55.        )
56.      ],
57.    ),
58.  ),
59. );
60. }
```

[Total: 20 Marks]

END OF EXAMINATION